```
entry
```

```
L0
LINENUMBER 84 L0
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getInterlace ()S
ICONST_1
IF_ICMPNE L1
```

```
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
BIPUSH 8
IF_ICMPGE L1
```

```
L1
LINENUMBER 88 L1
FRAME SAME
NEW optimizer/com/googlecode/pngtastic/core/PngImage
DUP
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.log : Loptimizer/com/googlecode/pngtastic/core/ILogger;
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngImage.<init> (Loptimizer/com/googlecode/pngtastic/core/ILogger;)V
ASTORE 4
L3
LINENUMBER 89 L3
ALOAD 4
ICONST_0
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.setInterlace (S)V
L4
LINENUMBER 91 L4
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getChunks ()Ljava/util/List;
INVOKEINTERFACE java/util/List.iterator () Ljava/util/Iterator;
ASTORE 5
L5
LINENUMBER 92 L5
ALOAD 0
ALOAD 4
ILOAD 2
ALOAD 5
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.processHeadChunks (Loptimizer/com/googlecode/pngtastic/core/PngImage;ZLjava/util/Iterator;)Loptimizer/com/googlecode/pngtastic/core/PngChunk;
ASTORE 6
L6
LINENUMBER 95 L6
ALOAD 0
ALOAD 6
ALOAD 5
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.getInflatedImageData (Loptimizer/com/googlecode/pngtastic/core/PngChunk;Ljava/util/Iterator;)Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;
ASTORE 7
L7
LINENUMBER 97 L7
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getWidth ()J
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
I2L
LMUL
L2F
LDC 8.0
FDIV
F2D
INVOKESTATIC java/lang/Math.ceil (D)D
D2I
ICONST_1
IADD
ISTORE 8
L8
LINENUMBER 99 L8
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getInterlace ()S
ICONST_1
IF_ICMPNE L9
```

```
L9
FRAME FULL [optimizer/com/googlecode/pngtastic/core/PngOptimizer optimizer/com/googlecode/pngtastic/core/PngImage I java/lang/Integer optimizer/com/googlecode/pngtastic/core/PngImage java/util/Iterator optimizer/com/googlecode/pngtastic/core/PngChunk optimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream I] []
LDC "a7c0bb15-ea53-49d8-a1bc-6fd256a068a8"
INVOKESTATIC edu/cmu/cs/mvelezce/tool/analysis/region/Regions.enter (Ljava/lang/String;)V
ALOAD 0
ALOAD 7
ALOAD 1
L12
LINENUMBER 101 L12
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
ILOAD 8
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getHeight ()J
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.getScanlines (Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;IIJIL java/util/List;
```

```
ALOAD 0
LDC "6ff09ce8-f261-443a-9c47-14f44b72f1f1"
INVOKESTATIC edu/cmu/cs/mvelezce/tool/analysis/region/Regions.enter (Ljava/lang/String;)V
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.pngInterlaceHandler : Loptimizer/com/googlecode/pngtastic/core/processing/PngInterlaceHandler;
ALOAD 1
L10
LINENUMBER 100 L10
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getWidth ()J
L2I
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getHeight ()J
L2I
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
ALOAD 7
INVOKEINTERFACE optimizer/com/googlecode/pngtastic/core/processing/PngInterlaceHandler.deInterlace (IIIILoptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;Ljava/util/List;
GOTO L11
```

```
L11
FRAME SAME1 java/util/List
ASTORE 9
L13
LINENUMBER 107 L13
NEW java/util/HashMap
DUP
INVOKESPECIAL java/util/HashMap.<init> ()V
ASTORE 10
L14
LINENUMBER 108 L14
INVOKESTATIC optimizer/com/googlecode/pngtastic/core/PngFilterType.standardValues ()[Loptimizer/com/googlecode/pngtastic/core/PngFilterType;
ASTORE 11
ALOAD 11
ARRAYLENGTH
ISTORE 12
ICONST_0
ISTORE 13
```

```
L2
LINENUMBER 85 L2
ALOAD 1
ARETURN
```

```
L15
FRAME FULL [optimizer/com/googlecode/pngtastic/core/PngOptimizer optimizer/com/googlecode/pngtastic/core/PngImage I java/lang/Integer optimizer/com/googlecode/pngtastic/core/PngImage java/util/Iterator optimizer/com/googlecode/pngtastic/core/PngChunk optimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream I java/util/List java/util/HashMap [Loptimizer/com/googlecode/pngtastic/core/PngFilterType; I I] []
ILOAD 13
ILOAD 12
IF_ICMPGE L16
```

```
L16
LINENUMBER 117 L16
FRAME SAME
ACONST_NULL
ASTORE 11
L22
LINENUMBER 118 L22
ACONST_NULL
ASTORE 12
L23
LINENUMBER 119 L23
ALOAD 10
INVOKEINTERFACE java/util/Map.entrySet () Ljava/util/Set;
INVOKEINTERFACE java/util/Set.iterator () Ljava/util/Iterator;
ASTORE 13
```

```
ALOAD 11
ILOAD 13
AALOAD
ASTORE 14
L17
LINENUMBER 109 L17
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.log : Loptimizer/com/googlecode/pngtastic/core/ILogger;
LDC "Applying filter: %s"
ICONST_1
ANEWARRAY java/lang/Object
DUP
ICONST_0
ALOAD 14
AASTORE
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/ILogger.debug (Ljava/lang/String;[Ljava/lang/Object;)V
L18
LINENUMBER 110 L18
ALOAD 0
ALOAD 9
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.copyScanlines (Ljava/util/List;)Ljava/util/List;
ASTORE 15
L19
LINENUMBER 111 L19
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.pngFilterHandler : Loptimizer/com/googlecode/pngtastic/core/processing/PngFilterHandler;
ALOAD 14
ALOAD 15
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
INVOKEINTERFACE optimizer/com/googlecode/pngtastic/core/processing/PngFilterHandler.applyFiltering (Loptimizer/com/googlecode/pngtastic/core/PngFilterType;Ljava/util/List;I)V
L20
LINENUMBER 113 L20
ALOAD 10
ALOAD 14
ALOAD 15
INVOKEINTERFACE java/util/Map.put (Ljava/lang/Object;Ljava/lang/Object;)Ljava/lang/Object;
POP
L21
LINENUMBER 108 L21
IINC 13 1
GOTO L15
```

```
L24
FRAME FULL [optimizer/com/googlecode/pngtastic/core/PngOptimizer optimizer/com/googlecode/pngtastic/core/PngImage I java/lang/Integer optimizer/com/googlecode/pngtastic/core/PngImage java/util/Iterator optimizer/com/googlecode/pngtastic/core/PngChunk optimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream I java/util/List java/util/HashMap optimizer/com/googlecode/pngtastic/core/PngFilterType [B java/util/Iterator] []
ALOAD 13
INVOKEINTERFACE java/util/Iterator.hasNext ()Z
IFEQ L25
```

```
ALOAD 13
INVOKEINTERFACE java/util/Iterator.next () Ljava/lang/Object;
CHECKCAST java/util/Map$Entry
ASTORE 14
L26
LINENUMBER 120 L26
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.pngCompressionHandler : Loptimizer/com/googlecode/pngtastic/core/processing/PngCompressionHandler;
ALOAD 0
ALOAD 14
INVOKEINTERFACE java/util/Map$Entry.getValue () Ljava/lang/Object;
CHECKCAST java/util/List
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.serialize (Ljava/util/List;)Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;
ALOAD 3
ICONST_0
INVOKEINTERFACE optimizer/com/googlecode/pngtastic/core/processing/PngCompressionHandler.deflate (Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;Ljava/lang/Integer;Z)[B
ASTORE 15
L27
LINENUMBER 121 L27
ALOAD 12
IFNULL L28
```

```
L25
LINENUMBER 128 L25
FRAME CHOP 2
ALOAD 0
ALOAD 9
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngOptimizer.copyScanlines (Ljava/util/List;)Ljava/util/List;
ASTORE 13
L31
LINENUMBER 129 L31
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.pngFilterHandler : Loptimizer/com/googlecode/pngtastic/core/processing/PngFilterHandler;
ALOAD 7
ALOAD 13
ALOAD 10
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getSampleBitCount ()I
INVOKEINTERFACE optimizer/com/googlecode/pngtastic/core/processing/PngFilterHandler.applyAdaptiveFiltering (Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;Ljava/util/List;Ljava/util/Map;I)V
L32
LINENUMBER 131 L32
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.pngCompressionHandler : Loptimizer/com/googlecode/pngtastic/core/processing/PngCompressionHandler;
ALOAD 7
ALOAD 3
ICONST_0
INVOKEINTERFACE optimizer/com/googlecode/pngtastic/core/processing/PngCompressionHandler.deflate (Loptimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream;Ljava/lang/Integer;Z)[B
ASTORE 14
L33
LINENUMBER 132 L33
ALOAD 0
GETFIELD optimizer/com/googlecode/pngtastic/core/PngOptimizer.log : Loptimizer/com/googlecode/pngtastic/core/ILogger;
LDC "Original=%d, Adaptive=%d, %s=%d"
ICONST_4
ANEWARRAY java/lang/Object
DUP
ICONST_0
ALOAD 1
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getImageData ()[B
ARRAYLENGTH
INVOKESTATIC java/lang/Integer.valueOf (I)Ljava/lang/Integer;
AASTORE
DUP
ICONST_1
ALOAD 14
ARRAYLENGTH
INVOKESTATIC java/lang/Integer.valueOf (I)Ljava/lang/Integer;
AASTORE
DUP
ICONST_2
ALOAD 11
AASTORE
DUP
ICONST_3
ALOAD 12
IFNONNULL L34
```

```
ALOAD 15
ARRAYLENGTH
ALOAD 12
ARRAYLENGTH
IF_ICMPGE L29
```

```
L34
FRAME FULL [optimizer/com/googlecode/pngtastic/core/PngOptimizer optimizer/com/googlecode/pngtastic/core/PngImage I java/lang/Integer optimizer/com/googlecode/pngtastic/core/PngImage java/util/Iterator optimizer/com/googlecode/pngtastic/core/PngChunk optimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream I java/util/List java/util/HashMap optimizer/com/googlecode/pngtastic/core/PngFilterType [B java/util/List [B] [optimizer/com/googlecode/pngtastic/core/Logger java/lang/String [Ljava/lang/Object; [Ljava/lang/Object; I]
ALOAD 12
ARRAYLENGTH
```

```
ICONST_0
GOTO L35
```

```
L28
LINENUMBER 122 L28
FRAME APPEND [java/util/Map$Entry [B]
ALOAD 15
ASTORE 12
L30
LINENUMBER 123 L30
ALOAD 14
INVOKEINTERFACE java/util/Map$Entry.getKey () Ljava/lang/Object;
CHECKCAST optimizer/com/googlecode/pngtastic/core/PngFilterType
ASTORE 11
```

```
L29
LINENUMBER 125 L29
FRAME SAME
GOTO L24
```

```
L35
LINENUMBER 133 L35
FRAME FULL [optimizer/com/googlecode/pngtastic/core/PngOptimizer optimizer/com/googlecode/pngtastic/core/PngImage I java/lang/Integer optimizer/com/googlecode/pngtastic/core/PngImage java/util/Iterator optimizer/com/googlecode/pngtastic/core/PngChunk optimizer/com/googlecode/pngtastic/core/processing/PngByteArrayOutputStream I java/util/List java/util/HashMap optimizer/com/googlecode/pngtastic/core/PngFilterType [B java/util/List [B] [optimizer/com/googlecode/pngtastic/core/Logger java/lang/String [Ljava/lang/Object; [Ljava/lang/Object; I I]
INVOKESTATIC java/lang/Integer.valueOf (I)Ljava/lang/Integer;
AASTORE
L36
LINENUMBER 132 L36
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/Logger.debug (Ljava/lang/String;[Ljava/lang/Object;)V
L37
LINENUMBER 135 L37
ALOAD 12
IFNULL L38
```

```
ALOAD 14
ARRAYLENGTH
ALOAD 12
ARRAYLENGTH
IF_ICMPGE L39
```

```
L38
LINENUMBER 136 L38
FRAME SAME
ALOAD 14
ASTORE 12
L40
LINENUMBER 137 L40
GETSTATIC optimizer/com/googlecode/pngtastic/core/PngFilterType.ADAPTIVE : Loptimizer/com/googlecode/pngtastic/core/PngFilterType;
ASTORE 11
```

```
L39
LINENUMBER 140 L39
FRAME SAME
NEW optimizer/com/googlecode/pngtastic/core/PngChunk
DUP
LDC "IDAT"
INVOKEVIRTUAL java/lang/String.getBytes ()[B
ALOAD 12
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngChunk.<init> ([B[B)V
ASTORE 15
L41
LINENUMBER 141 L41
ALOAD 4
ALOAD 15
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.addChunk (Loptimizer/com/googlecode/pngtastic/core/PngChunk;)V
```

```
L42
LINENUMBER 144 L42
FRAME APPEND [optimizer/com/googlecode/pngtastic/core/PngChunk]
ALOAD 6
IFNULL L43
```

```
L44
LINENUMBER 145 L44
ALOAD 6
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.isCritical () Z
IFEQ L45
```

```
L43
LINENUMBER 159 L43
FRAME SAME
ALOAD 4
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.getChunks () Ljava/util/List;
ASTORE 16
L54
LINENUMBER 160 L54
ALOAD 16
IFNULL L55
```

```
LDC "IDAT"
ALOAD 6
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.getTypeString () Ljava/lang/String;
INVOKEVIRTUAL java/lang/String.equals (Ljava/lang/Object;)Z
IFNE L45
```

```
LDC "IEND"
ALOAD 16
ALOAD 16
INVOKEINTERFACE java/util/List.size () I
ICONST_1
ISUB
INVOKEINTERFACE java/util/List.get (I)Ljava/lang/Object;
CHECKCAST optimizer/com/googlecode/pngtastic/core/PngChunk
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.getTypeString () Ljava/lang/String;
INVOKEVIRTUAL java/lang/String.equals (Ljava/lang/Object;)Z
IFNE L55
```

```
L46
LINENUMBER 146 L46
NEW java/io/ByteArrayOutputStream
DUP
ALOAD 6
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.getLength () I
INVOKESPECIAL java/io/ByteArrayOutputStream.<init> (I)V
ASTORE 16
L47
LINENUMBER 147 L47
NEW java/io/DataOutputStream
DUP
ALOAD 16
INVOKESPECIAL java/io/DataOutputStream.<init> (Ljava/io/OutputStream;)V
ASTORE 17
L48
LINENUMBER 149 L48
ALOAD 17
ALOAD 6
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.getData ()[B
INVOKEVIRTUAL java/io/DataOutputStream.write ([B)V
L49
LINENUMBER 150 L49
ALOAD 17
INVOKEVIRTUAL java/io/DataOutputStream.close ()V
L50
LINENUMBER 152 L50
NEW optimizer/com/googlecode/pngtastic/core/PngChunk
DUP
ALOAD 6
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngChunk.getType ()[B
ALOAD 16
INVOKEVIRTUAL java/io/ByteArrayOutputStream.toByteArray ()[B
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngChunk.<init> ([B[B)V
ASTORE 18
L51
LINENUMBER 153 L51
ALOAD 4
ALOAD 18
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.addChunk (Loptimizer/com/googlecode/pngtastic/core/PngChunk;)V
```

```
L56
LINENUMBER 161 L56
ALOAD 4
NEW optimizer/com/googlecode/pngtastic/core/PngChunk
DUP
LDC "IEND"
INVOKEVIRTUAL java/lang/String.getBytes ()[B
ICONST_0
NEWARRAY T_BYTE
INVOKESPECIAL optimizer/com/googlecode/pngtastic/core/PngChunk.<init> ([B[B)V
INVOKEVIRTUAL optimizer/com/googlecode/pngtastic/core/PngImage.addChunk (Loptimizer/com/googlecode/pngtastic/core/PngChunk;)V
```

```
L45
LINENUMBER 155 L45
FRAME SAME
ALOAD 5
INVOKEINTERFACE java/util/Iterator.hasNext () Z
IFEQ L52
```

```
L55
LINENUMBER 164 L55
FRAME APPEND [java/util/List]
LDC "a7c0bb15-ea53-49d8-a1bc-6fd256a068a8"
INVOKESTATIC edu/cmu/cs/mvelezce/tool/analysis/region/Regions.exit (Ljava/lang/String;)V
LDC "6ff09ce8-f261-443a-9c47-14f44b72f1f1"
INVOKESTATIC edu/cmu/cs/mvelezce/tool/analysis/region/Regions.exit (Ljava/lang/String;)V
ALOAD 4
ARETURN
```

```
exit
```

```
L52
FRAME SAME
ACONST_NULL
```

```
ALOAD 5
INVOKEINTERFACE java/util/Iterator.next () Ljava/lang/Object;
CHECKCAST optimizer/com/googlecode/pngtastic/core/PngChunk
GOTO L53
```

```
L53
FRAME SAME1 optimizer/com/googlecode/pngtastic/core/PngChunk
ASTORE 6
GOTO L42
```